



**DEPARTMENT OF ELECTRICAL AND INFORMATION
ENGINEERING**

**FACULTY OF ENGINEERING
UNIVERSITY OF RUHUNA**

**INDUSTRIAL TRAINING REPORT SUBMITTED IN PARTIAL FULFILLMENT
OF THE DEGREE OF BACHELOR OF THE SCIENCE OF ENGINEERING
ON**

09TH DECEMBER 2024

L.E. ROBOTICS (PVT) LTD.

(From 27.05.2024 - 18.08.2024)

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ACKNOWLEDGEMENT

First, I would like to express my heartfelt thanks to my parents for their constant support, guidance, and encouragement, which have shaped me into the person I am today and helped me pursue my university education.

I am very grateful to Senior Professor Kapila Jayasinghe, Director of Technical Operations at LE Robotics Pvt Ltd and Senior Professor at the University of Moratuwa, for his valuable guidance and supervision throughout my training. His support has greatly improved my technical knowledge and practical skills.

I would like to thank the management and staff of LE Robotics Pvt Ltd for their support, assistance, and for giving me this important training opportunity. Their willingness to share their knowledge and create a positive learning environment has been vital for my growth.

My sincere thanks go to the Engineering Education Center (EEC) at the Faculty of Engineering, University of Ruhuna, and NAITA for organizing this training program. I also appreciate the support and guidance from the lecturers in the Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna.

I would like to give special thanks to Engineer Saman Kumara and Technician Janaka Rathnayake for their guidance, advice, and support during my training period. Their knowledge and encouragement played an important role in my learning journey.

Finally, I would like to thank all the staff members of LE Robotics Pvt Ltd and everyone who was involved in organizing this training. This experience has given me practical knowledge, technical skills, and insights into robotics and electronics. I am truly thankful for this opportunity.

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EG/2021/4891

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Chapter 1: Introduction to the Training Establishment

1.1 Training Establishment - L. E. Robotics Pvt Ltd

L. E. Robotics Pvt Ltd is an emerging company, basically working in the field of robotics and automation. L. E. Robotics focuses on designing and developing futuristic robotic systems for industries like manufacturing, healthcare, and research. L. E. Robotics ensures the quality solutions to be high by adhering to the required industry standards. The systems are designed by L. E. Robotics using robotics, artificial intelligence, and machine learning to improve operational efficiency and reduce the extent of manual labor. It aims to create customized solutions between the company and its clients, thereby contributing towards the growth of the robotics sector.

The team at L. E. Robotics consists of highly skilled engineers and professionals who are always innovating and pushing the boundaries. The company also emphasizes environmental sustainability and corporate responsibility. Thus, fostering an environment of teamwork, integrity, and ethical practices, L. E. Robotics has earned its correct position in the partnerships to business interests interested in introducing robotic solutions to their operations. Figure 1 displays the official logo of LE Robotics Pvt Ltd, representing the company's brand and identity.



Figure 1: Logo of LE Robotics Pvt Ltd

1.2 History of the Company

Lanka Electronics (Pvt) Ltd was established in 1992 with the primary goal of advancing Sri Lanka's technology industry. Initially, the company developed low-cost traffic light systems tailored for Sri Lanka. It later expanded into designing laboratory-based electronic equipment and eventually moved to producing consumer electronics such as TV antennas and multi-plugs.

In 2010, the company began researching and developing electric bikes, marking a significant step into innovative product development. By 2015, Lanka Electronics ventured into robotics, starting various projects in this field. To better focus on robotics and automation solutions, the company launched a new subsidiary, LE Robotics, in 2018. This new entity was dedicated to providing automation solutions for industrial problems and conducting research and development in a well-organized manner.

LE Robotics prides itself on being the first in Sri Lanka to offer fully customizable robotics solutions designed and manufactured locally, catering to automation needs at affordable prices. By using locally developed electronic control systems and locally designed mechanical systems, the company ensures the highest level of customization for specific requirements. To maintain quality and reliability, LE Robotics incorporates world-class gearboxes, motors, and electronic components into its products. As a locally based solution provider, the company offers an approachable and flexible service unmatched by foreign providers, combining affordability with tailored solutions.

However, during the COVID-19 pandemic, the global shortage of integrated circuits (ICs) significantly affected the company. Many developments had to be halted, and the company was forced to reduce its workforce. Currently, it is gradually recovering and working towards regaining its growth trajectory.

1.3 Mission and Vision

Mission

Become the leader in affordable industrial robots, robot accessories and automation solutions using robotics in Sri Lanka and the Region.

Vision

To be the leading provider of innovative robotics and automation solutions in Sri Lanka, driving technological advancements, empowering industries, and fostering sustainable development for a smarter future.

1.4 Special Features of LE Robots

- Customizable to suit specific process requirements.
- Multiple operation modes to accommodate diverse motion and speed needs.
- Intuitive user interface with support for multiple devices.
- Easy access to spare parts for maintenance.
- Tailored electronic control systems to match organizational needs.
- High-quality design ensuring reliability

1.5 Services from LE Robotics

- **Process Analysis:** Identifying the automation needs and process challenges specific to your organization.
- **Product Customization:** Providing tailor-made solutions that best fit your requirements.
- **Local Expertise:** Offering expert knowledge and support with easy access.
- **Lifetime Support:** Ensuring continuous assistance and service throughout the lifecycle of the product.

1.6 Products from LE Robotics

1.6.1 6-DOF Robots

- Capable of positioning the end effector to a given x, y, z coordinates and orient along a given roll, pitch and yaw directions. Figure 2 illustrates the design and functionality of the Six DOF robot.
- Robotics solutions designed to mimic human arm movements.
- Fully customizable robotic movements to meet specific requirements.
- Approximately 1-meter reach and 0.1 mm placement precision.

Figure 3 provides detailed dimensions of the 6-DOF robot, showcasing its size and reach. Figure 5 presents the technical specifications and key features of the 6-DOF robot.



Figure 2: Six DOF Robot

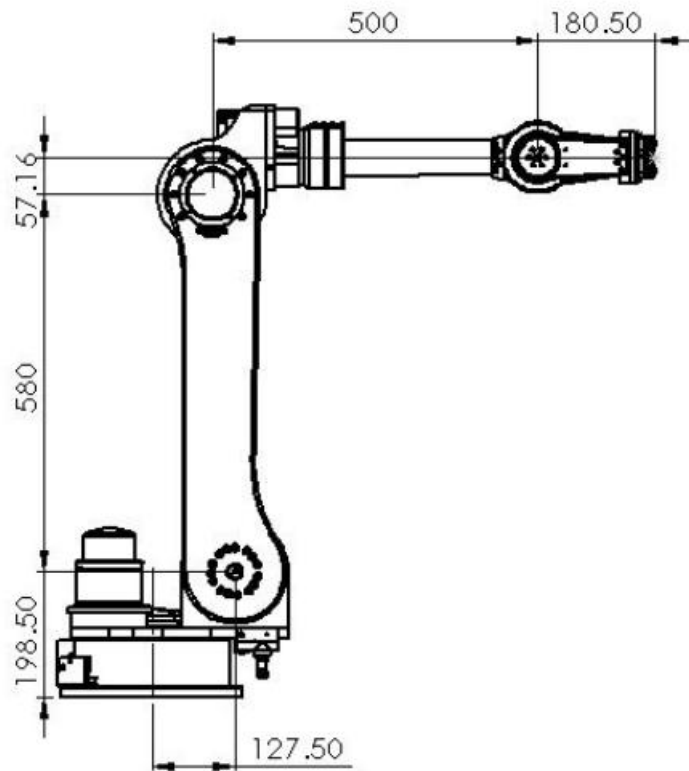


Figure 3: Dimensions of the 6-DOF Robot

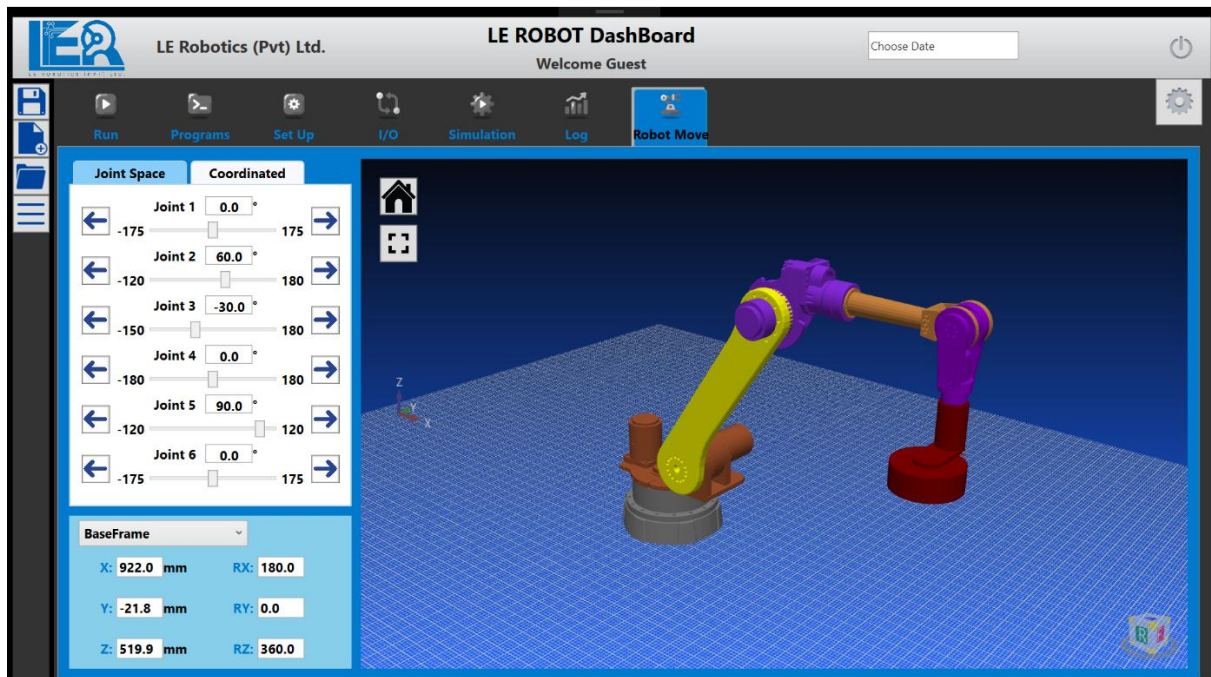


Figure 4: Dashboard of the 6-DOF Robot

Technical Data

Maximum reach	1388 mm
Maximum payload	12.5 kg
Rated payload	10 kg
Rated supplementary load, rotating column / link arm / arm	0 kg / 0 kg / 10 kg
Pose repeatability	0.1mm
Number of axes	6
Mounting position	Floor / Ceiling
Footprint	Ø 280 mm
Weight	approx. 75 kg

Figure 5: Technical Data of 6 DOF Robot

1.6.2 4-DOF Robots

- Popular option for small robotic assembly applications. Figure 6 illustrates the design and structure of the Four DOF robot.
- Robotics solutions offering high-speed performance, rigidity, and accuracy.
- Compact design with a variety of options.

- Around 500 mm reach and 0.1 mm placement precision. Figure 7 highlights the dimensions of the 4-DOF robot, including its reach and compact design and Figure 8 details the technical data and performance metrics of the 4-DOF robot.



Figure 6: Four DOF Robot

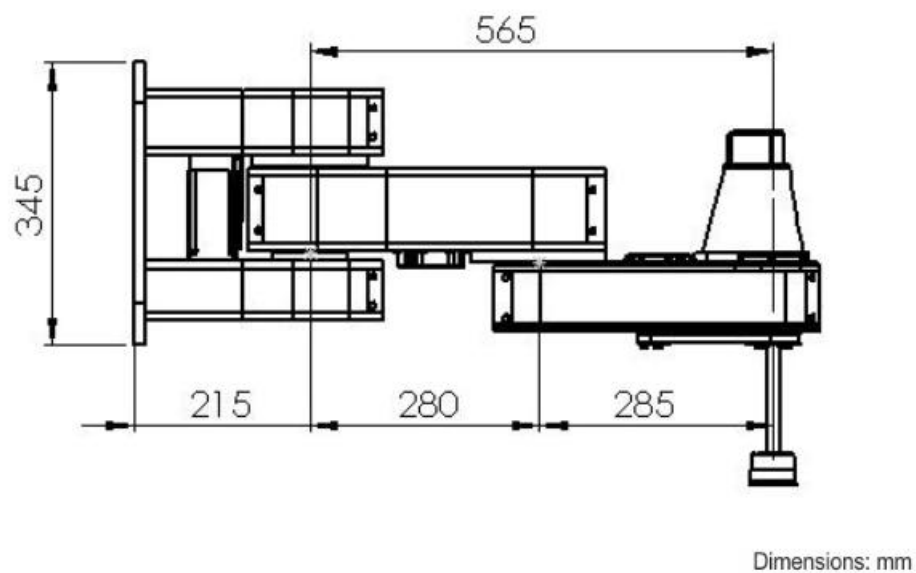


Figure 7: Dimensions of the 4-DOF Robot

Technical Data

Maximum reach	565 mm
Maximum payload	1 kg
Rated payload	0.5 kg
Pose repeatability	0.1mm
Number of axes	4
Mounting position	Wall
Footprint	345 mm x 304 mm
Weight	approx. 6 kg

Figure 8: Technical Data of 4 DOF Robot

1.6.3 Custom Made Robots

- Customizable reach and payload specifications based on specific process requirements.
- Robotics solutions designed for simple pick-and-place operations.

Chapter 2: Training Experiences- Technical

During my internship, I was assigned to the Research Center of LE Robotics (Pvt) Ltd. During this period, I was given a group project to develop a Compact Flash (CF) Card Reader. In addition to working on this project, I gained valuable hands-on experience in practical applications related to electronics and hardware description languages.

2.1 Compact Flash (CF) card

A Compact Flash (CF) card is a small, portable storage device widely used in digital cameras, industrial equipment, and embedded systems. It uses flash memory to store data and supports high-speed data transfer in a durable design. Figure 9 shows the physical structure and layout of a CompactFlash (CF) storage card.



Figure 9: CompactFlash (CF) storage card

(source- <https://shashinki.com/shop/used-sandisk-64gb-extreme-compactflash-memory-card-120mb-s-excellent-condition-p-59528.html>)

The CompactFlash (CF) storage card integrates a single-chip controller and flash memory modules within a compact. It features a 50-pin connector. The controller facilitates communication with a host system, enabling data to be read from and written to the flash memory modules. Figure 10 illustrates the block diagram of a CompactFlash card, explaining its components and data flow.

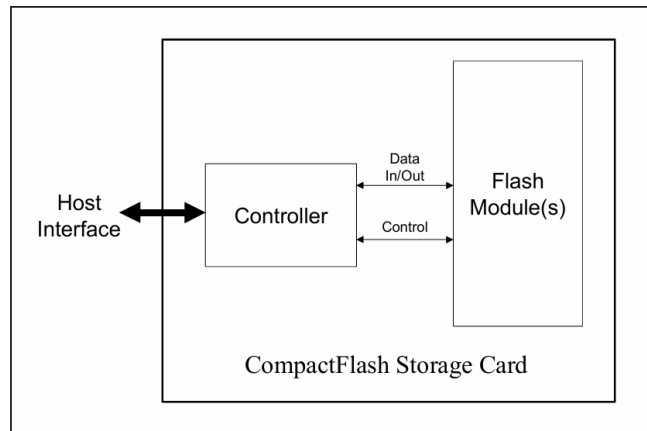


Figure 10: CompactFlash Storage Card Block Diagram

We designed a CF Card Reader board that enables communication between a CompactFlash storage card and a host system, allowing efficient data reading and writing operations. Figure 11 presents the printed circuit board (PCB) design for the CF card reader. Figure 12 shows the CF card reader board with a CompactFlash card inserted.

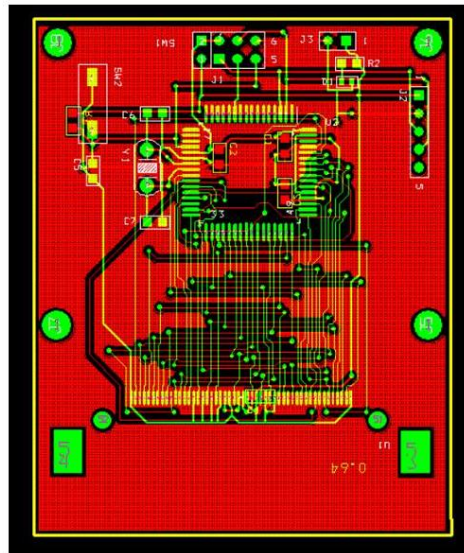


Figure 11: PCB Design of the CF Card Reader Board



Figure 12: CF Card Reader Board with CF Card

2.1.1 CF Card Modes

CompactFlash (CF) cards operate in three distinct modes, each suited for different applications:

- **PC Card ATA I/O Mode:** The CF card functions as an I/O device, allowing the host system to perform input/output operations. Typically used in applications that require interfacing with PCMCIA I/O devices.
- **PC Card ATA Memory Mode:** The CF card behaves like a standard memory device, accessed similarly to PCMCIA memory cards. Used for data storage and retrieval, particularly in embedded systems

- **True IDE Mode:** The CF card works like an IDE hard drive, making it compatible with ATA/IDE systems. This mode is turned on by connecting pin 9 of the CF card to the ground. It is required for CompactFlash cards but optional for CF+ cards.

2.1.2 Pin Assignments and Pin Type

CompactFlash (CF) cards use a 50-pin connector with pins assigned for specific purposes. These pins enable communication, power delivery, and control between the CF card and the host device. (Association, 2003) Table 1 lists the various pin types used in CompactFlash cards and their respective functions.

Table 1: Pin Types of CompactFlash (CF) Cards

Pin Types	Description
Power Pins	Supply power to the CF card.
Ground Pins	Provide a common ground connection.
Data Pins (Input/Output)	Facilitate data transfer between the card and the host.
Control Pins	Manage operations
Address Pins	Identify specific memory locations.

Control Pins in CompactFlash (CF) cards are specialized pins that manage the operation and communication between the card and the host system. These pins regulate key functions such as enabling the card, controlling data output, initiating writing, resetting the card, and managing interrupts.

Control Signals refer to the electrical signals sent through these control pins to instruct the CF card to perform specific tasks, such as reading, writing, or resetting. These signals help ensure the card operates correctly in different modes and synchronizes with the host system. Table 2 details the control signals in CompactFlash cards and their purposes.

Table 2: Control Signals in the CompactFlash (CF) Cards

Control Signal	Description
Chip Enable (CE1)	Activates the CF card. It is used to enable or disable the card's communication with the host system.
Output Enable (OE)	Enables output from the CF card to the host system. It controls when data can be read from the card.
Chip Select (CS0)	Selects the CF card in the memory space, enabling the card to communicate with the host.
Write Enable (WE)	Enables writing to the CF card. When active, the card can accept data for writing; when inactive, write operations are blocked.
Reset (RESET)	Resets the CF card, reinitializing it to a known state, often used during power-up or when a fault occurs.
Interrupt (INTRQ)	Sends an interrupt signal to the host system, notifying it of events such as data completion or errors.
I/O Control (I/O1, I/O2)	Manages the direction and flow of I/O operations, allowing the host to control data input/output.
Bus Width Enable (BWE)	Selects the data bus width (e.g., 8-bit, 16-bit) during communication between the CF card and the host.
Write Protect (WP)	Prevents writing to the card when active, protecting data from being overwritten.
Voltage Select (VSELECT)	Selects the operating voltage for the CF card (e.g., 3.3V or 5V), ensuring compatibility with the host system's voltage.
Data Ready (DRQ)	Indicates when data is ready to be transferred, signaling the host system to read the data.
Busy (-BUSY)	Indicates if the CF card is busy processing data, helping the host determine when it can perform further operations.

2.1.3 Registers in CompactFlash (CF) Cards

Registers in CompactFlash (CF) cards are small, specific memory locations that store data or control information used for communication between the card and the host system. These

registers are essential for managing operations such as reading, writing, configuration, and error handling. Table 3 provides an overview of the registers in CompactFlash cards and their specific roles.

Table 3: Registers in CompactFlash (CF) Cards

Register Name	Address	Description
Data Register	1F0h (offset 0, 8, 9)	Stores data that is read from or written to the CF card. Used for transferring data during read and write operations.
Error Register	1F1h (offset 1, 0Dh, Read-Only)	Contains error information and provides status about the success or failure of commands.
Feature Register	1F1h (offset 1, 0Dh, Write-Only)	Used to set specific features for the CF card, such as enabling special operations or modes.
Sector Count Register	1F2h (offset 2)	Holds the number of sectors to be read or written. Used with the Sector Number Register to define the range of data.
Sector Number Register	1F3h (offset 3)	Holds the starting sector number for read/write operations. Works with the Sector Count Register to specify the range of sectors involved.
Cylinder Registers	1F4h (offset 4) (Low Cylinder), 1F5h (offset 5) (High Cylinder)	Used in addressing sectors on the card, representing the high and low parts of the cylinder address.
Drive/Head Register	1F6h (offset 6)	Specifies the drive and head address for accessing data, commonly used in systems mimicking hard drives.

Status Register	1F7h (offset 7)	Provides the current status of the CF card, such as readiness to send/receive data or encountering errors.
Device Control Register	3F6h (offset Eh)	Controls various device functions, enabling or disabling specific features or operations on the CF card.
Identify Drive Register	1F8h–1FFh (various offsets)	Identifies the CF card and stores details like model number, serial number, and capabilities, similar to IDE hard drive identification.
Configuration Register	Base + 02h, 03h	Stores configuration options for the CF card, such as setting the operating modes (I/O, memory, True IDE).

2.2 FAT 32 File Structure

2.2.1 Structure

FAT32 (File Allocation Table 32) most often found on CompactFlash Cards, USB Drives and SD cards is the most used file system. Its design ensures that files are stored in an organized manner and that storage is easy to manage. The Boot Sector is the first sector in a FAT32 volume ID and it's never empty. It is used to give an overview of the file system such as its type, the clusters, the size of the clusters, and the location of most of its parts like the FAT and the Root Directory. After the Boot Sector, there are Reserved Sectors which are areas that are not empty but are set aside for system purposes and as a backup of the Boot Sector. (Stoffregen, 2005)

Following this is the File Allocation Table (FAT), which is another vital component of the file system. It is similar to a map. Its function is to indicate which clusters are free and which ones are already occupied, and how the occupied clusters are used to store files that span over a number of clusters. Following the last is the Root Directory which has information such as the names and sizes of files as well as the clusters in which the files begin stored.

With the only difference that FAT16 fixed the Root Directory, FAT32 uses a variable one that can expand in size as desired.

Finally, the Data Region occupies most of the storage space and contains the actual file and directory contents. Files are stored in clusters, with the FAT managing their allocation and sequence. The FAT32 structure also features redundancy by maintaining two copies of the FAT, ensuring data recovery in case of corruption. Its compatibility with various operating systems and support for large storage devices make FAT32 a widely adopted file system. Figure 13 depicts the structure of the FAT32 file system, outlining its key components.



Figure 13: FAT 32 File System Structure

2.2.2 Master Boot Record (MBR)

The drive's first sector, known as the Master Boot Record (MBR), is located at LBA = 0 and serves as a critical part of a storage device's structure. All IDE drives support sector access using Logical Block Addressing (LBA) and have a fixed sector size of 512 bytes.

The MBR layout:

- **Bootstrap Code:** The first 446 bytes contain the bootloader code Helps to boot the system.
- **Partition Table:** The next 64 bytes define the primary partitions on the drive.
- **Boot Signature:** The final 2 bytes are a signature value, always **0x55AA**.

Figure 14 explains the structure and components of the Master Boot Record (MBR).

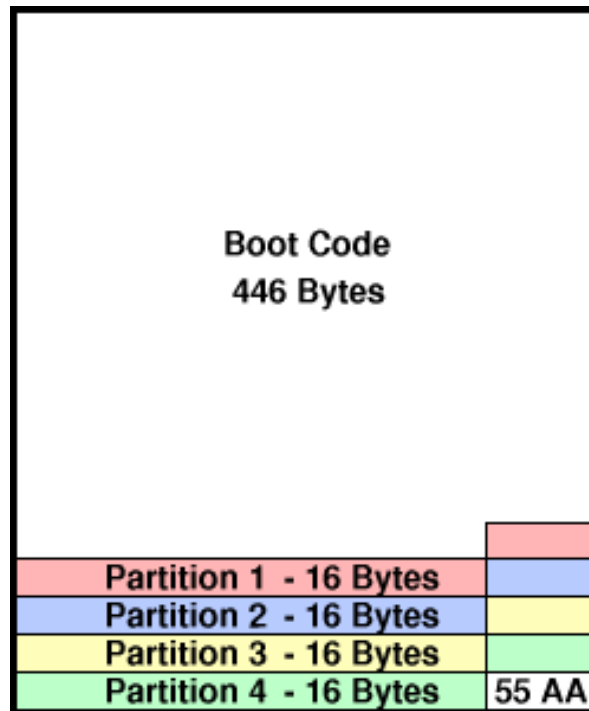


Figure 14: Master Boot Record (MBR)

2.2.3 Partition Information in the MBR

The **Master Boot Record (MBR)** can represent up to **four primary partitions**, with each partition described by a **16-byte entry** in the partition table. Figure 15 provides an example of a partition entry in the Master Boot Record and Table 4 outlines the FAT32 type codes and their corresponding meanings.

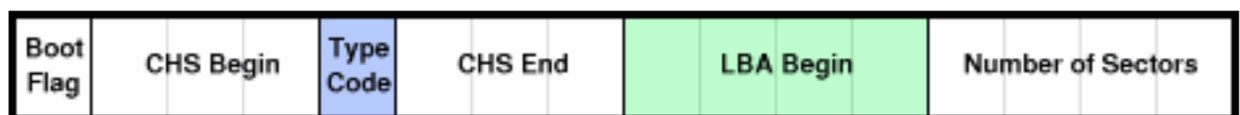


Figure 15: Partition Entry

Type Code: Tells what type of filesystem is supposed to be contained within the partition.

Table 4: FAT32 Type Codes and Their Meanings

Type Code (Hex)	Meaning	Description
0x0B	FAT32 (CHS)	FAT32 file system using Cylinder-Head-Sector (CHS) addressing.

0x0C	FAT32 (LBA)	FAT32 file system using Logical Block Addressing (LBA).
------	-------------	---

LBA Begin: It specifies the Logical Block Address (LBA) of the partition's first sector.

First, the Type Code of each partition entry in the MBR should be checked. It should be either **0x0B** or **0x0C**, as these are the two codes used for FAT32 partitions. Once the correct Type Code is identified, the LBA Begin address, located in bytes 9 to 12 of the partition entry, should be read. This address indicates where the FAT32 filesystem is located on the disk, pointing to the starting sector of the partition.

2.2.4 Volume ID

The first step for read the FAT32 file system is read the first sector and it known as the Volume ID. The LBA Begin address can found in the partition table. LBA Begin address is used to locate this sector. The Volume ID contains essential data about the physical layout of the FAT32 filesystem, such as the total number of sectors, the number of clusters, the size of each cluster, and other key filesystem parameters. This information is crucial for accessing and managing the filesystem's data structure. Figure 16 highlights the critical fields in the FAT32 Volume ID sector and Table 5 highlights the key fields in the FAT32 Volume ID sector, along with their functions.

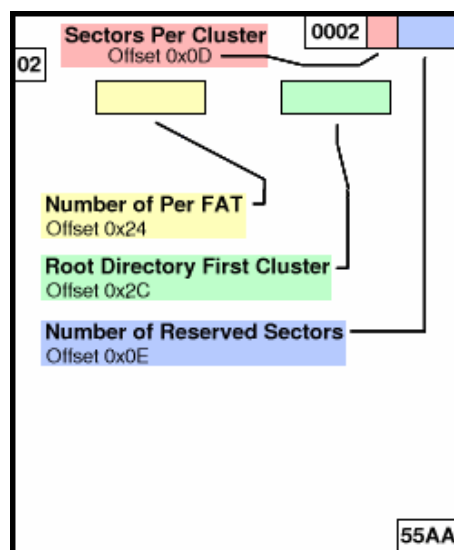


Figure 16: FAT32 Volume ID, Critical Fields

Table 5: Key Fields in the FAT32 Volume ID Sector

Field	Offset	Size	Value
Bytes Per Sector	0x0B	16 Bits	Always 512 Bytes
Sectors Per Cluster	0x0D	8 Bits	1, 2, 4, 8, 16, 32, 64, 128
Number of Reserved Sectors	0x0E	16 Bits	Usually, 0x20
Number of FATs	0x10	8 Bits	Always 2
Sectors Per FAT	0x24	32 Bits	Depends on disk size
Root Directory First Cluster	0x2C	32 Bits	Usually, 0x00000002
Signature	0x1FE	16 Bits	Always 0xAA55

Other Important Variables for Reading Fat-32:

- **Fat Begin LBA**
- **Cluster Begin LBA**

2.2.5 Arrangement of the FAT32 File System

The FAT32 filesystem begins with the **Volume ID**, located in the first sector. This is followed by an area known as **reserved sectors**, which is unused space designated for system purposes. After the reserved sectors, there are two copies of the **File Allocation Table (FAT)**, which store information about the allocation and usage of clusters. The rest of the filesystem is dedicated to storing **data**, organized into "clusters." Occasionally, a small portion of unused space may exist after the last cluster. Figure 17 illustrates the overall layout of the FAT32 filesystem, including its main regions.

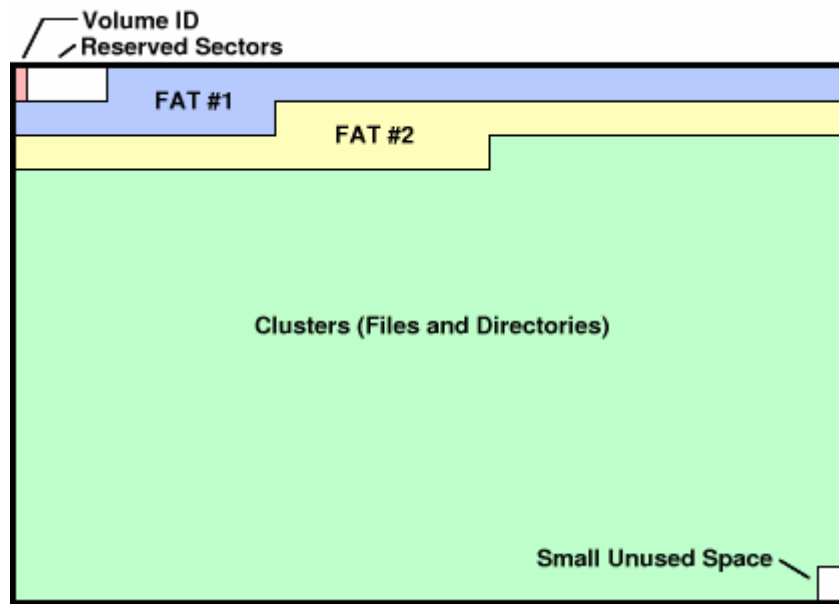


Figure 17: FAT32 Filesystem Overall Layout

The majority of the disk space in a FAT32 filesystem is allocated to the clusters section, which stores all files and directories. Cluster numbering starts at 2, meaning there are no clusters numbered 0 or 1. To access a specific cluster, the cluster number is converted into the Logical Block Address (LBA) of the IDE drive using a formula. This formula provides the exact location of the desired cluster on the disk.

Formula to convert cluster number into LBA address:

- **$LBA = (Cluster\ Number - 2) \times Sectors\ per\ Cluster + Cluster\ Start\ LBA$**

In the FAT32 filesystem, the root directory begins at the **first cluster** specified in the Volume ID. Reading the directory reveals the names and starting cluster locations of files and subdirectories. A critical point is that directories only provide the first cluster number of each file or subdirectory. To access additional clusters, the **FAT** (File Allocation Table) must be used. Initially, however, the directory structure is sufficient to locate where files and subdirectories begin.

Directory data is organized into **32-byte records**, with each sector containing exactly 16 records. This ensures no directory record spans across sector boundaries. The four types of directory records are:

1. **Normal Record with Short Filename** Contains file or directory information with a standard attribute.
2. **Long Filename Record** Used for filenames longer than the standard 8.3 format. These records have all four type bits set in the attribute field.
3. **Unused Record** Created when files are deleted. Identified by a first byte of **0xE5** and can be reused for new files.
4. **End of Directory Record** Indicates the end of the directory. Identified by a first byte of **0x00**.

Directory entries with a non-zero first byte (except **0xE5**) contain valid directory data. Their type is determined by checking the **attribute byte**. For practical purposes, short filenames (8.3 format) are sufficient to access files and subdirectories. FAT32 ensures all files have short names, even if users assign long names. This allows the system to ignore long filename records while accessing files.

This structure simplifies locating files, as directory records indicate where a file begins. Further clusters can then be accessed using the FAT. Table 6 describes the fields, offsets, and sizes within the 32-byte directory structure of the FAT32 file system.

Table 6: Fields, Offsets, and Sizes in the 32-Byte Directory Structure

Field	Offset	Size
Short Filename	0x00	11 Bytes
Attribute Byte	0x0B	8 Bits
First Cluster High	0x14	16 Bits
First Cluster Low	0x1A	16 Bits
File Size	0x1C	32 Bits

Figure 18 explains the 32-byte directory structure used for short filenames in the FAT32 file system.

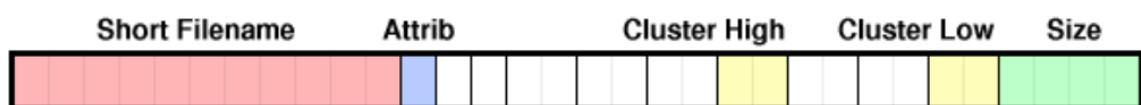


Figure 18: 32-Byte Directory Structure, Short Filename Format

The Attribute byte has six bits defined, as shown in the table below. Most simple firmware will check the Attribute byte to determine if the 32 bytes are a normal record or long filename data, and to determine if it is a normal file or a subdirectory. Long filename records have all four of the least significant bits set. Normal files rarely have any of these four-bit sets Table 7 explains the functions of each bit in the attribute byte used in directory records of FAT32.

Table 7: Attribute Byte Bit Functions in Directory Records

Attribute Bit	Function	LFN	Comment
0 (LSB)	Read Only	1	Should not allow writing
1	Hidden	1	Should not show in dir listing
2	System	1	File is an operating system
3	Volume ID	1	The filename is Volume ID
4	Directory	x	Is a subdirectory (32-byte records)
5	Archive	x	Has been changed since last backup
6	Unused	0	Should be zero
7 (MSB)	unused	0	Should be zero

2.2.6 File Allocation Table

Directory entries indicate the location of the first cluster of each file or subdirectory on the disk. The first cluster of the root directory is determined from the volume ID sector. For small files and directories that fit within a single cluster, the File Allocation Table (FAT) indicates that there are no additional clusters. Access to clusters beyond the first requires the use of the FAT, which in FAT32 consists of 32-bit entries. FAT16 and FAT12 use 16-bit and 12-bit entries, respectively.

The FAT is a large array of 32-bit integers, where each position corresponds to a cluster number, and the stored value indicates the next cluster in the file. Each sector of the FAT

contains 128 integers, facilitating the retrieval of the next cluster for a file. Bits 7-31 of the current cluster determine the sectors to read from the FAT, and bits 0-6 indicate which integer in that sector contains the next cluster number. If all bits are ones, it signifies the last cluster.

For files, the FAT does not specify the first cluster; this information is found in the directory, while the FAT is used for subsequent clusters. The number of clusters allocated matches the file size. A portion of the last cluster may remain unused unless the file size is an exact multiple of the cluster size. Zero-length files do not have any clusters, and their directory cluster number is zero. Files that fit within a single cluster will have the "FFFFFFFF" end-of-file marker at their FAT position. Figure 19 visualizes the File Allocation Table (FAT) and how it links cluster chains for files. (Dany, 2024)

Figure 19: File Allocation Table and Following Cluster Chains

2.3 Analyzing FAT32 Data Structures from a CompactFlash Card Example

Figure 20 provides a detailed look at the Master Boot Record (MBR) structure. Figure 21 illustrates the layout of a partition entry in the MBR.

2.3.1 Master Boot Record

```
struct MBR_Structure{
    unsigned char bootStrapCode[446];
    unsigned char partitionEntry_1[16];
    unsigned char partitionEntry_2[16];
    unsigned char partitionEntry_3[16];
    unsigned char partitionEntry_4[16];
    uint16_t      BootSignature;
};
```

Figure 20: MBR Structure

```
struct partition_entry_Structure {
    unsigned char CurrentStateOfPartition;
    unsigned char BeginningOfPartitionHead;
    uint16_t      BeginningOfPartitionSectorCylinder;
    unsigned char TypeOfPartition;
    unsigned char EndOfPartitionHead;
    uint16_t      EndOfPartitionSectorCylinder;
    uint32_t      NumberOfSectorsBetweenMBR_FirstSectorPartition;
    uint32_t      NumberOfSectorsPartition;
};
```

Figure 21: Partition Entry Structure

Figure 22 shows the detailed layout of the Master Boot Record of the example and Table 8 presents the data contained in the Master Boot Record (MBR).

Flash	EEPROM	Fuse Bits/Settings						
	00	01	02	03	04	05	06	07
000188	00	00	00	00	00	00	00	00
000190	00	00	00	00	00	00	00	00
000198	00	00	00	00	00	00	00	00
0001A0	00	00	00	00	00	00	00	00
0001A8	00	00	00	00	00	00	00	00
0001B0	00	00	00	00	00	00	00	00
0001B8	00	00	00	00	00	00	80	00
0001C0	03	00	0C	FA	FF	FB	FE	00
0001C8	00	00	72	39	BA	03	00	00
0001D0	00	00	00	00	00	00	00	00
0001D8	00	00	00	00	00	00	00	00
0001E0	00	00	00	00	00	00	00	00
0001E8	00	00	00	00	00	00	00	00
0001F0	00	00	00	00	00	00	00	00
0001F8	00	00	00	00	00	00	55	AA

Figure 22: Master Boot Record Layout

Table 8: Master Boot Record Data

Name	Color	Offset	Size(Bytes)
Boot Strap Code	No Color	0x000000	446
Partition Entry_1	Green	0x0001BE	16
Partition Entry_2	Purple	0x0001CE	16
Partition Entry_3	Blue	0x0001DE	16
Partition Entry_4	Red	0x0001EE	16
Boot Signature	Yellow	0x0001FE	2

Figure 23 highlights the specific details of Partition_1 entry in the MBR of the example and Table 9 provides detailed data for Partition Entry_1 within the Master Boot Record.

0001A8	00	00	00	00	00	00	00	00
0001B0	00	00	00	00	00	00	00	00
0001B8	00	00	00	00	00	00	80	00
0001C0	03	00	0C	FA	FF	FB	FE	00
0001C8	00	00	72	39	BA	03	00	00
0001D0	00	00	00	00	00	00	00	00
0001D8	00	00	00	00	00	00	00	00

Figure 23: Partition_1 Entry

Table 9: Partition Entry_1 Data

Name	Color	Offset	Size(Bytes)	Hex Value
Boot Flag	Green	0x0001BE	1	0x80
CHS Begin	Purple	0x0001BF	3	0x03
Type code	Blue	0x0001C2	1	0x0C
CHS END	Gray	0x0001C3	3	0xFBFFFA
LBA Begin	Red	0x0001C6	4	0xFE
Number of Sectors	Yellow	0x0001CA	4	0x03BA3972

2.3.2 Volume ID of the first partition.

Figure 24 presents the structure of the Boot Sector in a FAT32 file system and Figure 25 shows the Volume ID fields in the Boot Sector of the FAT32 file system. Figure 26 highlights important fields in the FAT32 Volume ID sector and Table 10 displays the critical fields in the FAT32 Volume ID with their corresponding data.

```
struct BS_Structure{
    unsigned char jumpBoot[3];
    unsigned char OEMName[8];
    unsigned int bytesPerSector;
    unsigned char sectorPerCluster;
    unsigned int reservedSectorCount;
    unsigned char numberOfFATs;
    unsigned int rootEntryCount;
    unsigned int totalSectors_F16;
    unsigned char mediaType;
    unsigned int FATsize_F16;
    unsigned int sectorsPerTrack;
    unsigned int numberOfHeads;
    unsigned long hiddenSectors;
    unsigned long totalSectors_F32;
    unsigned long FATsize_F32;
    unsigned int extFlags;
    unsigned int Version_of_fat32_driver;
    unsigned long ClusterNumberOfRoot;
    unsigned int sector_number_FSinfo;
    unsigned int sector_no_BackupBoot;
    unsigned char reserved[12];
    unsigned char driveNumber;
    unsigned char reserved1;
    unsigned char bootSignature;
    unsigned long volumeID;
    unsigned char volumeLabel[11];
    unsigned char FAT_name[8];
    unsigned char bootData[420];
    unsigned int bootEndSignature;
};
```

Figure 24: Boot Sector Structure

Flash	EEPROM	Fuse Bits/Settings						
	00	01	02	03	04	05	06	07
000000	EB	00	90	20	20	20	20	20
000008	20	20	20	00	02	40	20	00
000010	02	00	00	00	00	F8	00	00
000018	3F	00	FF	00	FE	00	00	00
000020	72	3D	BA	03	D1	1D	00	00
000028	00	00	00	00	02	00	00	00
000030	01	00	06	00	00	00	00	00
000038	00	00	00	00	00	00	00	00
000040	80	00	29	75	F4	E2	48	4E
000048	4F	20	4E	41	4D	45	20	20
000050	20	20	46	41	54	33	32	20
000058	20	20	00	00	00	00	00	00
000060	00	00	00	00	00	00	00	00
000068	00	00	00	00	00	00	00	00
000070	00	00	00	00	00	00	00	00

Figure 25: Volume ID (Boot Sector)

	00	01	02	03	04	05	06	07
000000	EB	00	90	20	20	20	20	20
000008	20	20	20	00	02	40	20	00
000010	02	00	00	00	00	F8	00	00
000018	3F	00	FF	00	FE	00	00	00
000020	72	39	BA	03	D1	1D	00	00
000028	00	00	00	00	02	00	00	00
000030	01	00	06	00	00	00	00	00
000038	00	00	00	00	00	00	00	00

Figure 26: Fat 32 Volume ID, Critical Fields

Table 10: Fat 32 Volume ID and Critical Fields Data

Field	Color	Offset	Size (Bytes)	Hex Value	Dec Value
Bytes Per Sector	Green	0x0B	2	0x200	512 Bytes
Sectors Per Cluster	Red	0x0D	1	0x40	64
Number of Reserved Sectors	Purple	0x0E	2	0x20	32
Number of FATs	Blue	0x10	1	0x02	2
Sectors Per FAT	Black	0x24	4	0x1D D1	7633
Root Directory First Cluster	Yellow	0x2C	4	0x02	2

2.3.3 Necessary Calculations

1) Fat Begin Address

Fat Begin Address	=	Partition Begin Address	+	Reserved Sector Count
--------------------------	----------	--------------------------------	----------	------------------------------

Partition Begin Address = 0xFE

Reserved Sector Count = 0x20

Fat Begin Address = 0xFE + 0x20

= 0x011E (Sector no = 286)

2) Cluster Begin Address

Cluster Begin Address	=	Fat Begin Address	+	(Number of FATs × FAT size)
------------------------------	----------	--------------------------	----------	------------------------------------

Fat Begin Address = 0x011E

Number of FATs = 0x02

FATsize_F32 = 0x1DD1

Cluster Begin Address = 0x011E + (0x02 × 0x1DD1)

= 0x3CC0 (Sector no = 15552)

3) Root Dir First Cluster Address

Root Directory First Cluster Address	=	Cluster Begin Address	+	[(Root Dir First Cluster) - 2] × (Sec Per Cluster)
---	----------	------------------------------	----------	---

Cluster Begin Address = 0x3CC0

Root Dir First Cluster = 0x02

Sectors Per Cluster = 0x40

Root Dir First Cluster Address = 0x3CC0 (Sector no = 15552)

Figure 27 illustrates the sector locations in the example.

Sector Address	0	1	MBR
	1		.
			.
			.
254	1		Volume ID
255	32		Res sec
286			
	7633		Fat 1
7919			
	7633		Fat 2
15552	64		Cluster 2
15616	64		Cluster 3
15680	64		Cluster 4
15744	64		.
15808	64		.
15872	64		.
15936	64		.
16000	64		.
.	64		Cluster 10
.			.
.			.
.			.
.			.
.			.

Figure 27: Sector Locations

The starting cluster of the Root Directory is Cluster Two (Starts from Sector 15552)

2.3.4 How to locate a file in the Root Directory

As an example, created ABC file in the CF card and Figure 28 shows the directory entry structure for the file ABC. Figure 29 details the 32-byte directory layout for files and folders.

Offset (d)	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	Decoded text
00007962768	6F	00	6F	00	73	00	74	00	50	00	00	00	65	00	72	00	o.o.s.t.P...e.r.
00007962784	E5	45	41	44	59	42	7E	31	54	4D	50	20	00	30	D4	42	âEADYB~lTMP .00B
00007962800	01	59	01	59	00	00	D5	42	01	59	05	00	00	00	80	00	.Y.Y..0B.Y....e.
00007962816	E5	43	00	46	00	2D	00	4D	00	43	00	0F	00	98	2E	00	âC.F.-.M.C...~..
00007962832	61	00	74	00	73	00	6C	00	6E	00	00	00	00	00	FF	FF	a.t.s.l.n....ÿÿ
00007962848	E5	46	2D	4D	43	7E	31	20	41	54	53	20	00	83	4A	7F	âF-MC~l ATS .fJ.
00007962864	0B	59	0E	59	00	00	37	AF	08	59	06	00	C5	03	00	00	.Y.Y..7~.Y..Â...
00007962880	E5	41	54	5F	33	32	20	20	48	20	20	10	34	69	81		âAT_32 H .4i.
00007962896	0B	59	0E	59	00	00	C6	42	07	59	07	00	63	03	00	00	.Y.Y..EB.Y.c...
00007962912	E5	2E	00	76	00	73	00	00	00	FF	FF	0F	00	EC	FF	FF	â..v.s...ÿÿ..lÿÿ
00007962928	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ..ÿÿÿÿ
00007962944	E5	53	34	44	41	35	7E	31	20	20	12	00	7B	73	81		âS4DA5~l ...{s.
00007962960	0B	59	0B	59	00	00	74	81	0B	59	08	00	00	80	00	00	.Y.Y..t...e.e..
00007962976	E5	6D	00	65	00	6E	00	74	00	2E	00	0F	00	9F	74	00	âm.e.n.t....ÿt.
00007962992	78	00	74	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF	x.t...ÿÿÿÿ..ÿÿÿÿ
00007963008	E5	4E	00	65	00	77	00	20	00	54	00	0F	00	9F	65	00	âN.e.w. .T...ÿe.
00007963024	78	00	74	00	20	00	44	00	6F	00	00	00	63	00	75	00	x.t. .D.o...c.u.
00007963040	E5	45	57	54	45	58	7E	31	54	58	54	20	00	53	5C	55	âEWTEX~lTXT .S\U
00007963056	0E	59	0E	59	00	00	5D	55	0E	59	00	00	00	00	00	00	.Y.Y..]U.Y.....
00007963072	41	42	43	20	20	20	20	20	54	58	54	20	18	53	5C	55	ABC TXT .S\U
00007963088	0E	59	0E	59	00	00	6C	55	0E	59	08	00	10	00	00	00	.Y.Y..lU.Y.....
00007963104	E5	6D	00	65	00	6E	00	74	00	2E	00	0F	00	9F	74	00	âm.e.n.t....ÿt.
00007963120	78	00	74	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF	x.t...ÿÿÿÿ..ÿÿÿÿ
00007963136	E5	4E	00	65	00	77	00	20	00	54	00	0F	00	9F	65	00	âN.e.w. .T...ÿe.
00007963152	78	00	74	00	20	00	44	00	6F	00	00	00	63	00	75	00	x.t. .D.o...c.u.
00007963168	E5	45	57	54	45	58	7E	31	54	58	54	20	00	C1	17	59	âEWTEX~lTXT .Â.Y
00007963184	0E	59	0E	59	00	00	18	59	0E	59	00	00	00	00	00	00	.Y.Y...Y.Y.....

Figure 28: Directory Entry of the File ABC

41 42 43 20 20 20 20 20 54 58 54 20 18 53 5C 55 | ABC TXT .S\U
0E 59 0E 59 00 00 6C 55 0E 59 08 00 10 00 00 00 | .Y.Y..1U.Y.....

Figure 29: 32-byte Directory Layout

Table 11 shows the directory entry data for the file ABC, detailing its attributes and locations.

Table 11: Data of the Directory Entry of the File ABC

Field	Offset	Size	Color	Value
Short Filename	0x00	11 Bytes	Red	ABC TXT
Attribute Byte	0x0B	8 Bits	Purple	32
First Cluster High	0x14	16 Bits	Green	0x0000
First Cluster Low	0x1A	16 Bits	Blue	0x0008
File Size	0x1C	32 Bits	Yellow	0x00000010

High order value: 00 00 -> little-endian to big-endian -> 00 00

Low-order value: 08 00 -> little-endian to big-endian -> 00 08

Cluster number = 00 00 00 08 = 8 (decimal)

Therefore, the first cluster where the file contents can be found is cluster 8. Cluster 8 starts from Sector 15936. Figure 30 illustrates the location of Cluster 8 in the FAT32 file system.

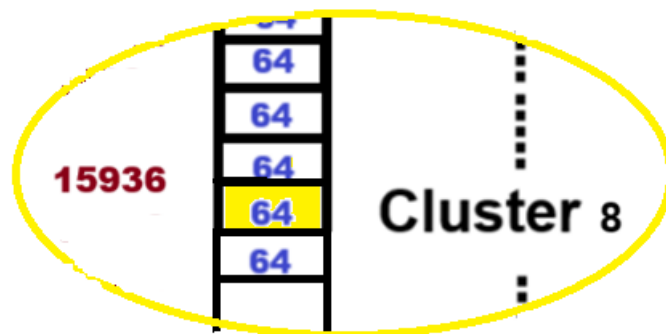


Figure 30: Location of Cluster 8

Figure 31 displays the contents of the first sector of the file ABC.

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text	
0007C8000	74	68	69	73	61	72	61	20	64	68	61	74	68	69	79	61	thisara dhathiya	Sector 15,936
0007C8010	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C80F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8100	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8110	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8120	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8130	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8140	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8150	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8170	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C81F0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8200	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		Sector 15,937
0007C8210	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8220	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0007C8230	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		

Figure 31: First Sector of the File ABC

2.3.5 How to locate a Sub-Directory in the Root Directory

Figure 32 shows the file and folder structure of a CompactFlash card.

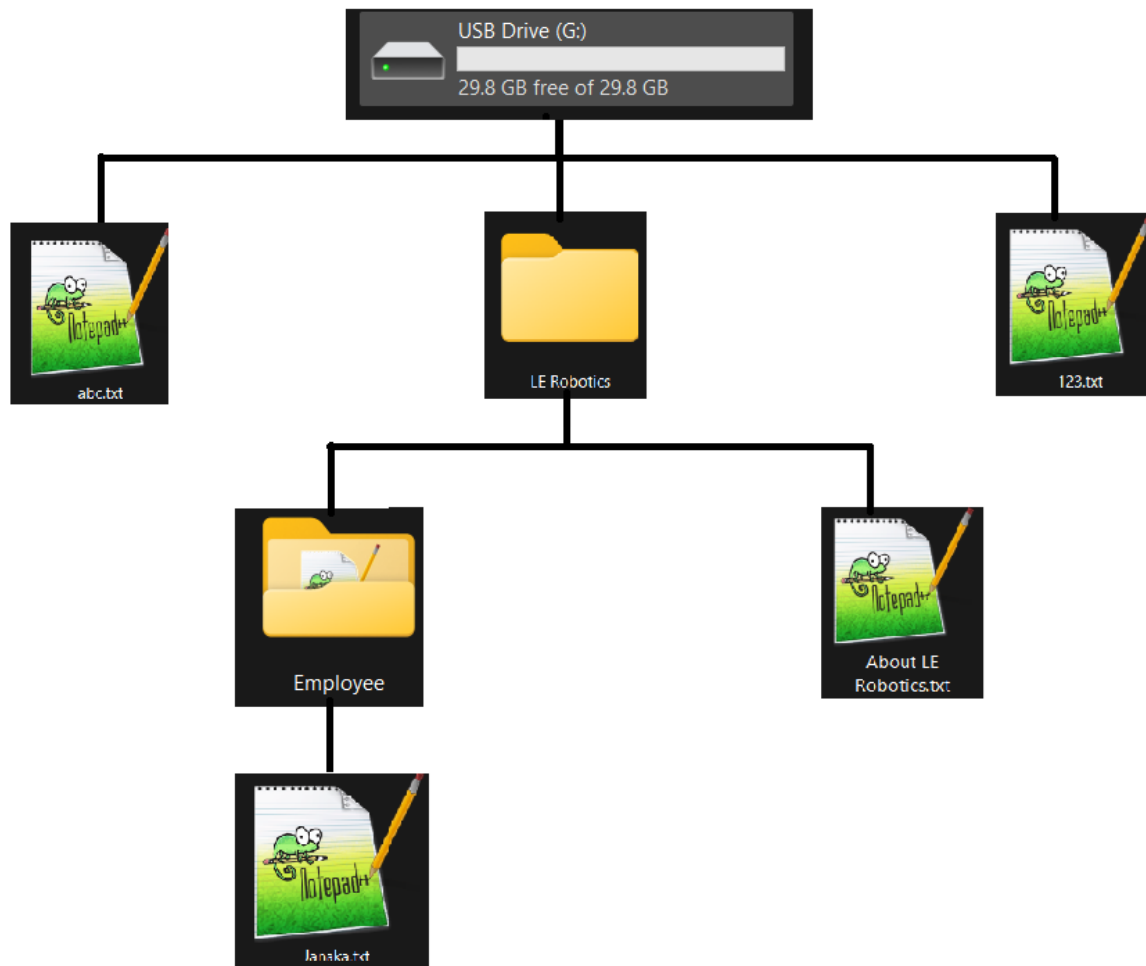


Figure 32: Files and Folders in CF Card

Figure 33 provides the directory entry for the folder 'LE Robotics' located in the root directory. Figure 34 details the layout of the directory entry for the 'LE Robotics' folder and Table 12 describe the directory entry data for the folder 'LE Robotics' in the root directory.

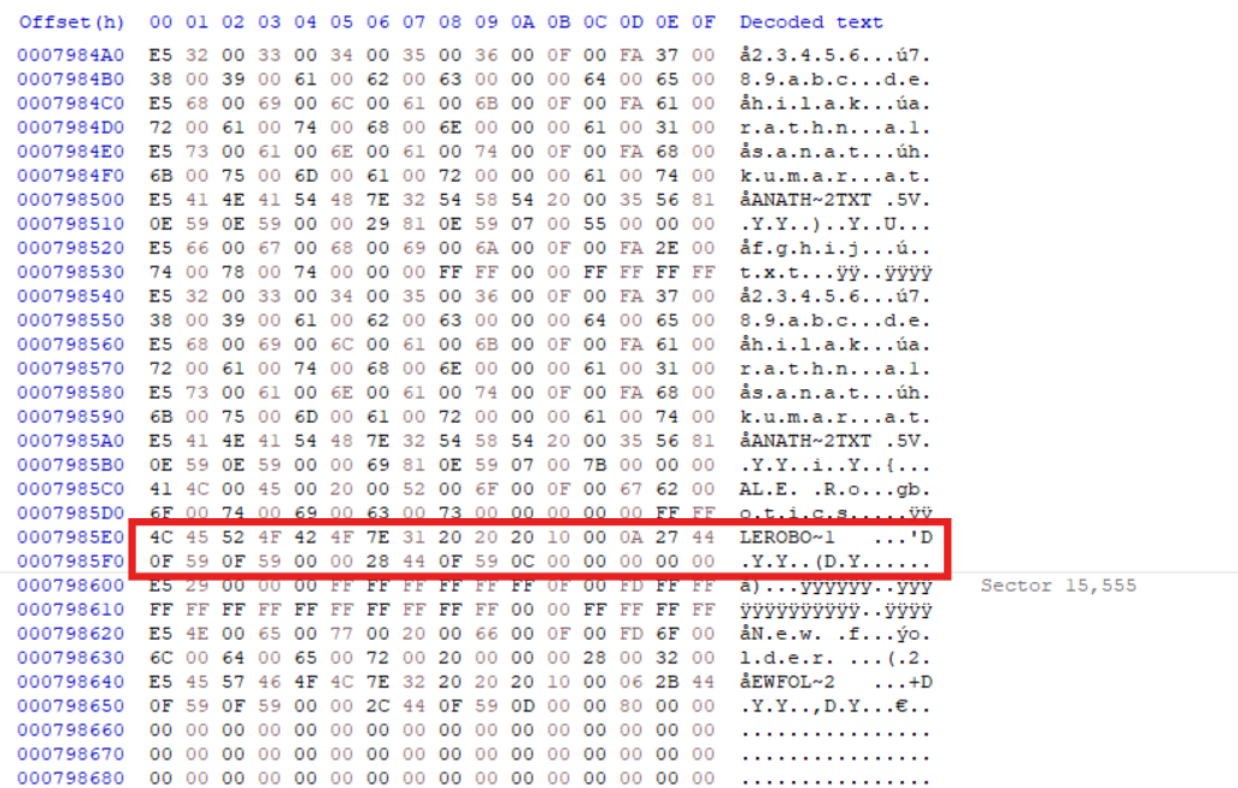


Figure 33: Directory Entry of the Folder "LE Robotics" in the Root Directory

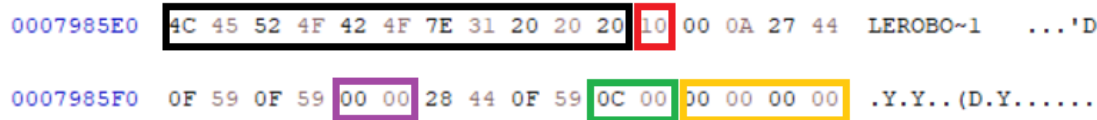


Figure 34: Layout of the Directory Entry of the Folder "LE Robotics"

Table 12: Data of the Directory Entry of the Folder "LE Robotics"

Field	Offset	Size	Color	Value
Short Filename	0x00	11 Bytes	Black	LEROBO~1
Attribute Byte	0x0B	8 Bits	Red	16
First Cluster High	0x14	16 Bits	Purple	0x0000

First Cluster Low	0x1A	16 Bits	Green	0x000C
File Size	0x1C	32 Bits	Yellow	0x00000000

High order value: 00 00 -> little-endian to big-endian -> 00 00

Low-order value: 0C 00 -> little-endian to big-endian -> 00 0C

Cluster number = 00 00 00 0C = 12 (decimal)

Therefore, the first cluster where the file contents can be found is cluster 12. Cluster 12 starts from Sector 16192. Figure 35 shows the subdirectory of the folder 'LE Robotics'

```

Offset(h)  00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
0007E8000  2E 20 20 20 20 20 20 20 20 20 20 10 00 0A 27 44  .      ...'D      Sector 16,192
0007E8010  0F 59 0F 59 00 00 28 44 0F 59 0C 00 00 00 00 00  .Y.Y..(D.Y.....
0007E8020  2E 2E 20 20 20 20 20 20 20 20 20 10 00 0A 27 44  ..      ...'D
0007E8030  0F 59 0F 59 00 00 28 44 0F 59 00 00 00 00 00 00  .Y.Y..(D.Y.....
0007E8040  E5 49 00 6E 00 74 00 65 00 72 00 0F 00 48 6E 00  âI.n.t.e.r...Hn.
0007E8050  00 00 FF FF FF FF FF FF FF 00 00 FF FF FF FF  ..YYYYYYYY..YYYY
0007E8060  E5 4E 54 45 52 4E 20 20 20 20 20 10 00 0C 42 44  âNTERN      ...BD
0007E8070  0F 59 0F 59 00 00 43 44 0F 59 0D 00 00 80 00 00  .Y.Y..CD.Y...€..
0007E8080  E5 4E 00 65 00 77 00 20 00 66 00 0F 00 DD 6F 00  âN.e.w. .f...Ÿo.
0007E8090  6C 00 64 00 65 00 72 00 00 00 00 00 FF FF FF FF  l.d.e.r....YYYY
0007E80A0  E5 45 57 46 4F 4C 7E 31 20 20 20 10 00 88 4C 44  âEWFOL~1  ..^LD
0007E80B0  0F 59 0F 59 00 00 4D 44 0F 59 0E 00 00 80 00 00  .Y.Y..MD.Y...€..
0007E80C0  41 45 00 6D 00 70 00 6C 00 6F 00 0F 00 F8 79 00  AE.m.p.l.o...øy.
0007E80D0  65 00 65 00 00 00 FF FF FF FF 00 00 FF FF FF FF  e.e...YYYY..YYYY
0007E80E0  45 4D 50 4C 4F 59 45 45 20 20 20 10 00 12 51 44  EMPLOYEE      ...QD
0007E80F0  0F 59 0F 59 00 00 52 44 0F 59 0E 00 00 00 00 00  .Y.Y..RD.Y.....
0007E8100  42 74 00 69 00 63 00 73 00 2E 00 0F 00 AE 74 00  Bt.i.c.s.....@t.
0007E8110  78 00 74 00 00 00 FF FF FF FF 00 00 FF FF FF FF  x.t...YYYY..YYYY
0007E8120  01 41 00 62 00 6F 00 75 00 74 00 0F 00 AE 20 00  .A.b.o.u.t...@ .
0007E8130  4C 00 45 00 20 00 52 00 6F 00 00 00 62 00 6F 00  L.E. .R.o...b.o.
0007E8140  41 42 4F 55 54 4C 7E 31 54 58 54 20 00 24 5B 44  ABOUTL~1TXT .$(D
0007E8150  0F 59 0F 59 00 00 2B 46 0F 59 0D 00 A9 02 00 00  .Y.Y..+F.Y...@...
0007E8160  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E8170  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E8180  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E8190  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0007E81F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....

```

Figure 35: Sub Directory of the Folder LE Robotics

2.3.6 How to locate a file in the Sub-Directory in the Root Directory

Figure 36 illustrates the directory entry for the file 'About LE Robotics' in its subdirectory. Figure 37 explains the layout of the directory entry for the file 'About LE Robotics' and Table 13 highlights the directory entry data of the folder 'LE Robotics,' showing its specific fields and attributes.

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text		
0007E8000	2E	20	20	20	20	20	20	20	20	20	20	10	00	0A	27	44	.	...D	Sector 16,192
0007E8010	0F	59	0F	59	00	00	28	44	0F	59	0C	00	00	00	00	00	.Y.Y..(D.Y.....		
0007E8020	2E	2E	20	20	20	20	20	20	20	20	20	10	00	0A	27	44	..	...D	
0007E8030	0F	59	0F	59	00	00	28	44	0F	59	00	00	00	00	00	00	.Y.Y..(D.Y.....		
0007E8040	E5	49	00	6E	00	74	00	65	00	72	00	0F	00	48	6E	00	âI.n.t.e.r...Hn.		
0007E8050	00	00	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	..YYYYYYYY..YYYY		
0007E8060	E5	4E	54	45	52	4E	20	20	20	20	20	10	00	0C	42	44	âNTERN	...BD	
0007E8070	0F	59	0F	59	00	00	43	44	0F	59	0D	00	00	80	00	00	.Y.Y..CD.Y...€..		
0007E8080	E5	4E	00	65	00	77	00	20	00	66	00	0F	00	DD	6F	00	âN.e.w. .f...Ÿo.		
0007E8090	6C	00	64	00	65	00	72	00	00	00	00	00	00	FF	FF	FF	l.d.e.r.....YYYY		
0007E80A0	E5	45	57	46	4F	4C	7E	31	20	20	20	10	00	88	4C	44	âEWFOl~1	...LD	
0007E80B0	0F	59	0F	59	00	00	4D	44	0F	59	0E	00	00	80	00	00	.Y.Y..MD.Y...€..		
0007E80C0	41	45	00	6D	00	70	00	6C	00	6F	00	0F	00	F8	79	00	AE.m.p.l.o...ey.		
0007E80D0	65	00	65	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF	e.e...YYYY..YYYY		
0007E80E0	45	4D	50	4C	4F	59	45	45	20	20	20	10	00	12	51	44	EMPLOYEE	...QD	
0007E80F0	0F	59	0F	59	00	00	52	44	0F	59	0E	00	00	00	00	00	.Y.Y..RD.Y.....		
0007E8100	42	74	00	69	00	63	00	73	00	2E	00	0F	00	AE	74	00	Bt.i.c.s.....@t.		
0007E8110	78	00	74	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF	x.t...YYYY..YYYY		
0007E8120	01	41	00	62	00	6F	00	75	00	74	00	0F	00	AE	20	00	.A.b.o.u.t...@.		
0007E8130	4C	00	45	00	20	00	52	00	6F	00	00	00	62	00	6F	00	L.E. .R.o...b.o.		
0007E8140	41	42	4F	55	54	4C	7E	31	54	58	54	20	00	24	5B	44	ABOUTL~1TXT	.\$[D	
0007E8150	0F	59	0F	59	00	00	2B	46	0F	59	0D	00	A9	02	00	00	.Y.Y...+F.Y...@...		
0007E8160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			
0007E8170	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			
0007E8180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			
0007E8190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			
0007E81A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			
0007E81B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00			

Figure 36: Sub Directory Entry of the File “About LE Robotics”

0007E8140	41	42	4F	55	54	4C	7E	31	54	58	54	20	00	24	5B	44	ABOUTL~1TXT	.\$[D
0007E8150	0F	59	0F	59	00	00	2B	46	0F	59	0D	00	A9	02	00	00	.Y.Y...+F.Y...@...	

Figure 37: Layout of the Directory Entry of the File “About LE Robotics”

Table 13: Data of the Directory Entry of the Folder "LE Robotics"

Field	Offset	Size	Color	Value
Short Filename	0x00	11 Bytes	Black	ABOUTL~1TXT
Attribute Byte	0x0B	8 Bits	Red	32
First Cluster High	0x14	16 Bits	Purple	0x0000
First Cluster Low	0x1A	16 Bits	Green	0x000D
File Size	0x1C	32 Bits	Yellow	681 bytes

High order value: 00 00 -> little-endian to big-endian -> 00 00

Low-order value: 0C 00 -> little-endian to big-endian -> 00 0D

Cluster number = 00 00 00 0D = 13 (decimal)

Therefore, the first cluster where the file contents can be found is cluster 13. Cluster 13 starts from Sector 16256, Figure 38 displays the contents of the first sector of the file 'About LE Robotics.

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text	
0007F0000	57	65	20	61	72	65	20	74	68	65	20	66	69	72	73	74	We are the first	Sector 16,256
0007F0010	20	69	6E	20	74	68	65	20	53	72	69	20	4C	61	6E	6B	in the Sri Lank	
0007F0020	61	6E	20	6D	61	72	6B	65	74	20	74	6F	20	6F	66	66	an market to off	
0007F0030	65	72	20	61	20	66	75	6C	6C	79	20	63	75	73	74	6F	er a fully custo	
0007F0040	6D	69	73	61	62	6C	65	20	72	6F	62	6F	74	69	63	73	misable robotics	
0007F0050	20	73	6F	6C	75	74	69	6F	6E	73	20	6D	61	64	65	20	solutions made	
0007F0060	69	6E	20	53	72	69	20	4C	61	6E	6B	61	20	66	6F	72	in Sri Lanka for	
0007F0070	20	79	6F	75	72	20	61	75	74	6F	6D	61	74	69	6F	6E	your automation	
0007F0080	20	6E	65	65	64	73	20	66	6F	72	20	61	6E	20	61	66	needs for an af	
0007F0090	66	6F	72	64	61	62	6C	65	20	70	72	69	63	65	20	77	fordable price w	
0007F00A0	69	74	68	20	74	68	65	20	65	78	70	65	72	74	69	73	ith the expertis	
0007F00B0	65	20	62	61	73	65	64	20	69	6E	20	53	72	69	20	4C	e based in Sri L	
0007F00C0	61	6E	6B	61	2E	0D	0A	57	69	74	68	20	74	68	65	20	anka...With the	
0007F00D0	6C	6F	63	61	6C	6C	79	20	64	65	76	65	6C	6F	70	65	locally develop	
0007F00E0	64	20	65	6C	65	63	74	72	6F	6E	69	63	20	63	6F	6E	d electronic con	
0007F00F0	74	72	6F	6C	20	73	79	73	74	65	6D	73	20	61	6E	64	trol systems and	
0007F0100	20	6C	6F	63	61	6C	6C	79	20	64	65	73	69	67	6E	65	locally designe	
0007F0110	64	20	61	6E	64	20	6D	61	6E	75	66	61	63	74	75	72	d and manufactur	
0007F0120	65	64	20	6D	65	63	68	61	6E	69	63	61	6C	20	73	79	ed mechanical sy	
0007F0130	73	74	65	6D	73	2C	20	77	65	20	63	61	6E	20	61	73	stems, we can as	
0007F0140	73	75	72	65	20	79	6F	75	20	74	6F	20	70	72	6F	76	sure you to prov	
0007F0150	69	64	65	20	74	68	65	20	62	65	73	74	20	63	75	73	ide the best cus	
0007F0160	74	6F	6D	69	73	61	74	69	6F	6E	20	66	6F	72	20	79	tomisation for y	
0007F0170	6F	75	72	20	73	70	65	63	69	66	69	63	20	6E	65	65	our specific nee	
0007F0180	64	2E	0D	0A	57	6F	72	6C	64	E2	80	93	43	6C	61	73	d...Worldâ€“Clas	
0007F0190	73	20	71	75	61	6C	69	74	79	20	67	65	61	72	20	62	s quality gear b	
0007F01A0	6F	78	65	73	2C	20	6D	6F	74	6F	72	73	20	61	6E	64	oxes, motors and	
0007F01B0	20	65	6C	65	63	74	72	6F	6E	69	63	20	63	6F	6D	70	electronic comp	
0007F01C0	6F	6E	65	6E	74	73	20	77	69	6C	6C	20	61	73	73	75	onents will assu	
0007F01D0	72	65	20	74	68	65	20	71	75	61	6C	69	74	79	20	61	re the quality a	
0007F01E0	6E	64	20	72	65	6C	69	61	62	69	6C	69	74	79	20	6F	nd reliability o	
0007F01F0	66	20	6F	75	72	20	70	72	6F	64	75	63	74	73	2E	0D	f our products..	
0007F0200	0A	42	65	69	6E	67	20	61	20	6C	6F	63	61	6C	6C	79	.Being a locally	Sector 16,257
0007F0210	20	62	61	73	65	64	20	73	6F	6C	75	74	69	6F	6E	20	based solution	
0007F0220	70	72	6F	76	69	64	65	72	2C	20	77	65	20	61	69	6D	provider, we aim	
0007F0230	20	74	6F	20	70	72	6F	76	69	64	65	20	61	6E	20	65	to provide an e	
0007F0240	61	73	69	6C	79	20	61	70	70	72	6F	61	63	68	61	62	asily approachab	
0007F0250	6C	65	20	61	6E	64	20	66	6C	65	78	69	62	6C	65	20	le and flexible	
0007F0260	73	65	72	76	69	63	65	20	61	74	20	61	6E	20	61	66	service at an af	
0007F0270	66	6F	72	64	61	62	6C	65	20	70	72	69	63	65	20	75	fordable price u	
0007F0280	6E	6D	61	74	63	68	65	64	20	77	69	74	68	20	66	6F	nmatched with fo	

Figure 38: First Sector of the File “About LE Robotics”

2.3.7 Analysis of Data Transfer Speed Based on Oscilloscope Observations

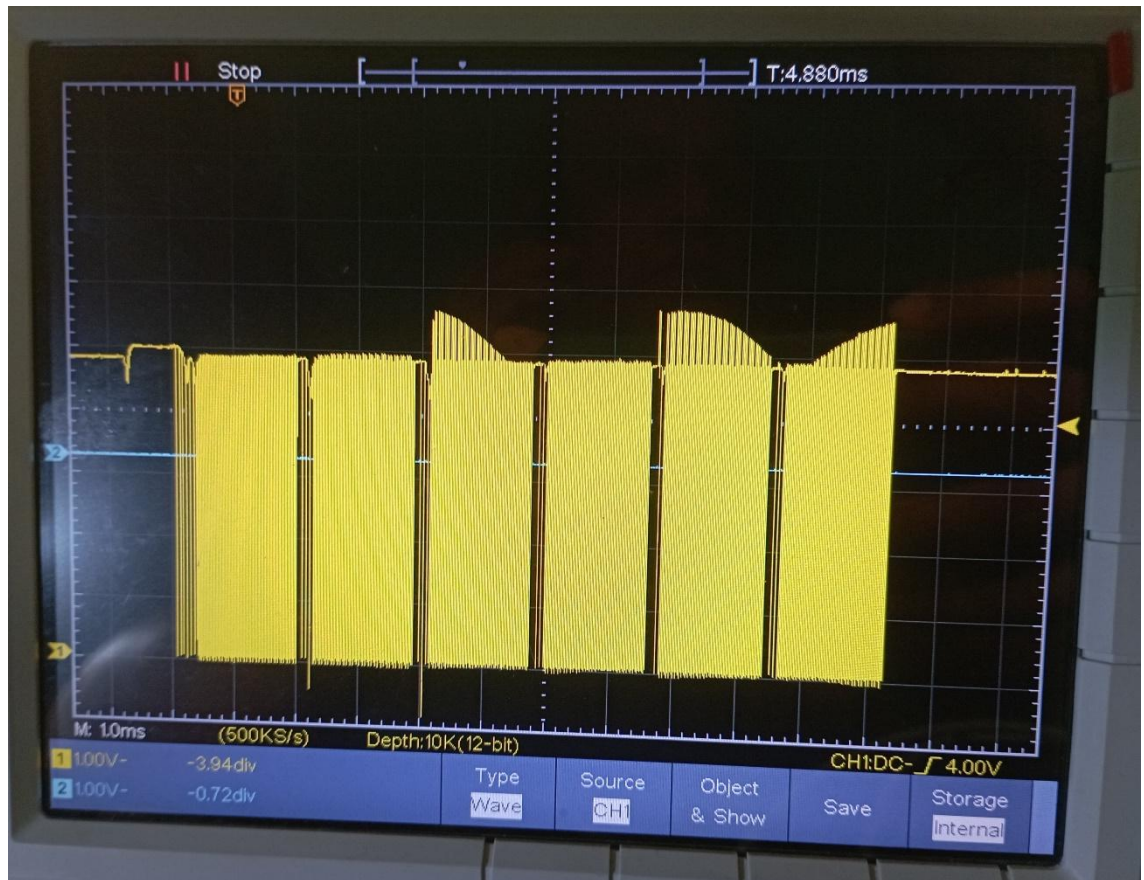


Figure 39: Observed Output from Oscilloscope

Figure 39 shows the output observed on an oscilloscope during the testing phase. According to the Figure 39, reading speed is approximately 275 Kbytes/Sec.

Chapter 3: Training Experiences (Management)

3.1 Management Structure

Le Robotics (Pvt) Ltd is a startup company and it has a streamlined management structure. Le Robotics (Pvt) Ltd focus on research and development. As an affiliated institution of its parent company, Lanka Electronics (Pvt) Ltd, it benefits from shared expertise and resources while maintaining operational independence. The company is led by a Managing Director who oversees key functions.

- Director Human Resources
- Director Technical Operations
- Director Marketing
- Director Finance

In addition to these key roles, the company is supported by different departments that work together to keep operations running smoothly, such as:

- Research and Development (R&D)
- Quality Control (QC)
- Production
- Stores
- Suppliers
- Account
- Sales
- M/R

Le Robotics is well-positioned to achieve its research objectives and contribute to technological advancements in the field. Figure 40 depicts the organizational management structure of LE Robotics Pvt Ltd.

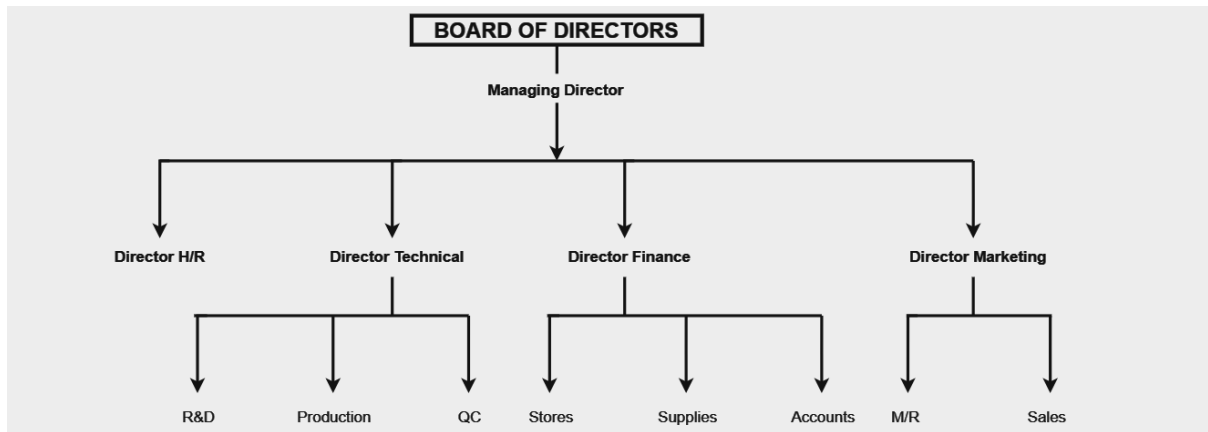


Figure 40: Management Structure

At LE Robotics Pvt Ltd, effective project management and teamwork are key to ensuring the successful completion of tasks. When a new project is received, the company holds an initial meeting to discuss its scope, divide relevant parts of the project among team members, and develop a time plan with clear milestones. This approach ensures that all members are aligned with the project's objectives and timelines.

Daily meetings play a crucial role in maintaining progress and addressing challenges. Each morning, before work begins at 7:45 am, and each evening, after work concludes around 6:30 pm, the company conducts short meetings. These sessions focus on reviewing progress, updating plans, and solving problems collaboratively under the guidance of the technical director.

Additionally, every Monday morning and Friday evening, special weekly meetings are held. These meetings evaluate the achievements of the past week and plan the workloads for the upcoming week. They also serve as an opportunity to address any broader issues that may impact the team's performance.

In case of significant challenges or urgent issues within a project, team members can request special meetings with the technical director. These meetings allow all members to share their insights and suggestions, fostering an environment of collaboration and problem-solving.

This structured approach to project management not only ensures efficiency but also encourages teamwork and open communication, enabling the successful execution of projects.

Chapter 4: Professional Standards and Engineering Ethics

As engineers and professionals, it is our responsibility to respect and adhere to Professional Standards and Engineering Ethics. When designing, creating, and maintaining systems and products that may impact public safety, health, and welfare, following these standards is essential. The IEEE Code of Ethics serves as a guide for engineers to make ethical decisions and encourages ethical behavior across all aspects of their work.

As a research and development company, we handle a significant amount of confidential information. During training, we may work on various projects, but we must ensure that confidential information is not shared outside the company. It's not just technical information that needs protection, but also other sensitive data, as market competition could make some information valuable to competitors. At the start of the training, we signed a confidentiality agreement, and below are the key points of this agreement:

Confidential Information:

- Disclosure of the company's "Trade Secrets" to external parties is strictly prohibited. This includes proprietary, financial, and business-related data.
- The company owns these secrets and has implemented measures to prevent their disclosure to the public.

Confidentiality During and After Training:

- All confidential information must be kept secure during training and after it ends.
- Reasonable precautions must be taken to prevent the exposure of the company's trade secrets and confidential data.

Return of Confidential Information:

- All confidential property or information must be returned to the company at the end of the training period.

Violation Consequences:

- Any breach of the confidentiality agreement may lead to legal actions, such as injunctions, restraining orders, and claims for financial damages caused by the breach.

Prohibition on External Projects:

- During the training period, involvement in subcontracted projects or employment with other organizations that could interfere with the company's interests is prohibited.

Legal Action for Breach:

- In the case of a breach of any part of the confidentiality agreement, the company has the right to seek legal remedies, including an injunction to prevent further violations and recover any financial losses.

As engineers working at the R&D center of Le Robotics Pvt Ltd, the workday starts at 8:00 am, and punctuality is expected. As a research center, there are projects that need to be completed within specific timeframes. If delays occur due to a lack of knowledge or other reasons, additional time may be required. However, we should always aim to complete projects on time, as this is a good practice for engineers.

In addition, I had another responsibility related to key control in one section. I had to check and ensure that all doors and windows were closed, and all power buttons and switches were turned off before the section was closed. Sometimes, I double-checked these items as part of a good practice.

Chapter 5: Environment and Sustainability

LE Robotics Pvt Ltd uses energy-efficient machinery and equipment in its production processes. The company utilizes energy-efficient soldering equipment and an LED lighting system to illuminate the workplace. Additionally, the workplace is designed to maximize natural sunlight, reducing the need for artificial lighting. As an electronics and robotics company, LE Robotics Pvt Ltd has an effective waste management system. The company recycles electronic components and materials wherever possible. When disposing of hazardous waste, such as batteries or chemicals, they strictly follow proper environmental regulations.

At the workplace, the company implements the 5S concept (Sort, Set in Order, Shine, Standardize, Sustain), which significantly improves efficiency and productivity. Figure 41 illustrates the 5S philosophy for workplace organization and efficiency.

Sort: Removing unnecessary tools and materials from the workplace reduces clutter and creates more space for efficient operations.

Set in Order: Organizing tools and equipment systematically ensures easy access, saving time during production processes.

Shine: Regular cleaning and maintenance of machinery improve equipment performance and prevent breakdowns.

Standardize: Establishing clear workflows and procedures ensures consistency in production and reduces errors.

Sustain: Continuously practicing 5S fosters a culture of discipline and continuous improvement, enhancing overall operational efficiency. (government, n.d.)



Figure 41: The 5S Philosophy

(Source-Anton,2024)

By adopting 5S, LE Robotics optimizes resource utilization, reduces waste, and creates a safer and more organized workplace. The company also provides training to employees and trainees, helping them understand the importance of sustainability and encouraging the adoption of eco-friendly practices in their work.

Chapter 6: Summary and Conclusions

6.1 Summary

As a trainee engineer, I had a great opportunity to work at LE Robotics Pvt Ltd. During my training period, I was supervised by Senior Professor Kapila Jayasinghe, who is the Director of Technical Operations at LE Robotics Pvt Ltd and a senior professor at the University of Moratuwa.

During this time, I developed technical knowledge in the field of electronics and gained valuable practical experience. I worked on group projects, which helped me develop important group working qualities such as collaboration and teamwork. Additionally, I tested some PCBs with identified issues and developed state diagrams for specific cases. These tasks significantly enhanced my technical skills and understanding.

LE Robotics Pvt Ltd provides robotics and automation solutions to global electronics manufacturing companies. I had the valuable opportunity to visit some of these companies, which greatly enhanced my exposure and knowledge in the field. During these visits, I gained insights into the global electronics industry, the challenges it faces, the design and optimization of production lines, opportunities available for electrical and electronic engineers, and emerging trends shaping the future of the electronics field.

6.2 Conclusions

My industrial training at LE Robotics Pvt Ltd. was an excellent learning platform that allowed me to bridge the gap between academic knowledge and practical applications. The exposure to real-world engineering challenges helped me enhance my technical and problem-solving skills significantly. Tasks like testing of PCBs, creation of state diagrams, and participation in group projects contributed to my growth as an aspiring engineer.

Visits to international companies manufacturing electrical equipment impressed me and gave valuable ideas about problems, chances, and perspectives of development in the industry. These experiences inspired me to be more resourceful in innovative solutions and to consider the role an electrical or electronic engineer will play to push forward technology.

Overall, I can describe my training as excellent since it helped me not only develop technical skills but also teamwork, communication, and organization skills. However, it would be great if the accessibility of more advanced tools or resources could be increased for further improvement in the learning process among trainees.

I strongly believe that the present industrial training that EEC and NAITA have adopted is very helpful for students, but if emerging technologies and world best practices are emphasized, it will be more useful. Undoubtedly, this training has prepared me for challenges in the future that lie ahead in the engineering field.

List of Abbreviations

CF	-	CompactFlash
DOF	-	Degrees of Freedom
IEEE	-	Institute of Electrical and Electronics Engineers
NAITA	-	National Apprenticeship and Industrial Training Authority
EEC	-	Engineering Education Center
PLC	-	Programmable Logic Controller
PID	-	Proportional Integral Derivative
QC	-	Quality Control
R&D	-	Research and Development
FAT	-	File Allocation Table
Dir	-	Directory
LBA	-	Logical Block Address
USB	-	Universal Serial Bus
CE1	-	Chip Enable
OE	-	Output Enable
CS0	-	Chip Select
WE	-	Write Enable
RESET	-	Reset
INTRQ	-	Interrupt Request
BWE	-	Bus Width Enable
WP	-	Write Protect
VSELECT	-	Voltage Select
DRQ	-	Data Ready
BUSY	-	Busy
ATA	-	Advanced Technology Attachment
IDE	-	Integrated Drive Electronics

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